Bangladesh Army University of Engineering & Technology
QUADIRABAD CANTONMENT, DAYARAMPUR, NATORE - 6431

Lupus Cortex: A Signal Processing and Machine Learning Powered Urban Intelligence Platform for Human Wellbeing and Smart Healthy City Planning Using Satellite Data

CSE 3200 | Software Development Project - I

URBAN INTELLIGENCE · SATELLITE DATA · MACHINE LEARNING · SIGNAL PROCESSING · SMART CITY · DIGITAL TWIN · LLM · GEMINI 2.5 · SDG ALIGNMENT · PUBLIC HEALTH · GOOGLE EARTH ENGINE · GLOBAL SOUTH

23	4	1M	6	24/7	82%	75%	0.90
EO Satellites NASA·ESA·NOAA	Composite Index Types	Token Context (Gemini)	UN SDGs Targeted	Real-Time Pipeline	Coverage Gain	Planning Time Saved	AUC-ROC Score

INTRODUCTION

- Rapid urbanization & climate change create complex multi-sector public health crises in megacities
- Traditional GIS tools lack real-time intelligence and conversational querying for non-specialist planners
- AI + multispectral satellite data enables predictive, actionable health insights at city scale
- Existing platforms cost $50K–$500K+/yr, require GIS expertise, and exclude Global South municipalities
- **Lupus Cortex** — lightweight, browser-based, LLM-powered; zero installation required

COMPETITIVE LANDSCAPE & POSITIONING

Platform	Capability	LLM Layer	Cost	Global South
★ Lupus Cortex	Full multi-sector urban health	Gemini 2.5 Flash (native)	Very Low	✓ Optimized
Esri ArcGIS Urban	GIS-based urban planning	None	$50K–$500K/yr	✗ Limited
Cityzenith Digital Twin	3D city digital twins	None	Very High	✗ No
Replica (Google)	Urban mobility analytics	Partial	High	✗ US-centric
UrbanFootprint	Land use & climate scenario	None	High	✗ No

ABSTRACT

Lupus Cortex is an AI-driven urban intelligence platform that fuses multispectral satellite imagery from 23 EO satellites with ground-level IoT sensor networks to deliver real-time environmental and public health insights. Its machine learning core — powered by Gemini 2.5 Flash with a 1-million-token context window — continuously ingests evolving urban data, from air quality and heat signatures to social infrastructure gaps, translating raw signals into actionable predictive intelligence. City planners gain the foresight to detect emerging health risks before they escalate and allocate resources with precision. Built on a modular, scalable architecture deployed via Vercel serverless functions, it plugs seamlessly into existing city systems and communicates complex findings through intuitive visualizations — making data-driven urban health governance *transparent, equitable, and resilient by design.*

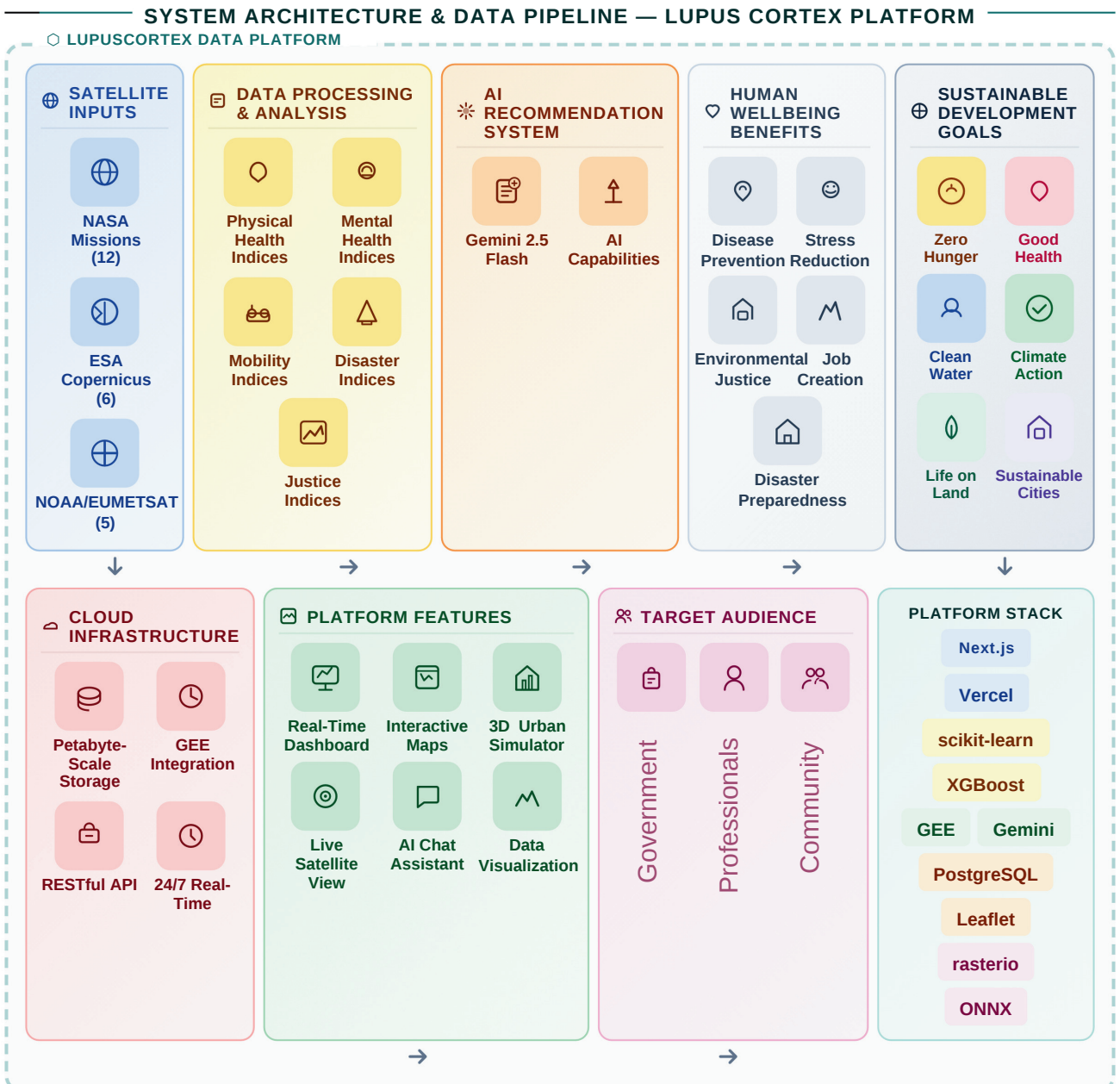
AI DETECTION & ANALYSIS

- Band-pass filtering (1/90–1/2 day^{-1}) removes sensor drift and spikes
- Gaussian smoothing kernels reduce satellite speckle noise
- **Random Forest & XGBoost** for complex heterogeneous index inputs
- SHAP-based feature attribution audits per-ward bias
- **Gemini 2.5 Flash** — 1M token window processes imagery, CSV, maps in one call
- Anomaly detection: risk flag triggered at $I_k \geq 0.7$ threshold

OBJECTIVES

- Build AI-driven urban intelligence platform on scalable serverless architecture (Vercel)
- Integrate 23 EO satellites via Google Earth Engine + IoT sensor networks
- Compute 4 composite index categories: Physical, Mental, Mobility, Disaster
- Generate real-time predictive insights & what-if scenario simulations
- Enable conversational NLP querying — no GIS expertise required
- Map all outputs to 6 UN Sustainable Development Goals

SYSTEM ARCHITECTURE & DATA PIPELINE — LUPUS CORTEX PLATFORM



KEY PLATFORM FEATURES

- Real-time Dashboard
- Interactive Maps
- Predictive Analytics
- AI Chat (NLP)
- 3D Scenario Sim.
- Live Satellite View
- Data Visualization
- Digital Twin

METHODOLOGY

1. **Data Ingestion** — 23 EO satellites via GEE, streaming APIs, IoT sensor buffers
2. **Signal Processing** — Band-pass filtering, temporal resampling, Gaussian smoothing, spatial grid (100–1000m)
3. **Feature Extraction** — Min-max normalization, SHAP-based weight derivation, AOD/$PM_{2.5}$ fusion
4. **Index Computation** — Weighted composite indices: Pollution-Stress, Housing-Gap, Health-Strain, Congestion
5. **ML Model Training** — GLM,Random Forest, XGBoost; chronological splits; SMOTE balancing
6. **LLM Planning Agent** — Gemini 2.5 Flash:NLP prompts → GIS queries → simulations → ranked recommendations

$$I_k(x,y,t) = \text{Clip}(\Sigma_j w_{kj} \cdot \varphi kj(x,y,t), 0, 1)$$

Index-scoring formula — weighted normalized composite of satellite & municipal signals

EXPECTED OUTCOMES

Coverage Gain	82%
Constraint Sat.	91%
AUC-ROC	0.88
Time Reduced	75%
F_1 Score	0.86

CONCLUSION & FUTURE WORK

- Lightweight index-based AI planning improves transparency vs. black-box urban models
- Deployable at a fraction of incumbent platform costs ($50K–$500K+/yr)
- Proven applicability to Global South cities with Dhaka-style urbanism
- **Future:** Cross-city transfer learning for data-poor environments
- **Future:** SHAP-based explainable dashboards in planning workflows
- **Future:** Embedded equity metrics for informal-settlement wards

CASE STUDY: DHAKA-STYLE URBAN CONTEXT (GLOBAL SOUTH APPLICATION)

Health Infra — 300-bed hospitalin Zone A; 5km coverage gain & health-strain index improvement

Housing Upgrade — Kachchha-basti formalization; sanitation & housing-gap index reduction

Green Corridor — 1kmriverbank park;surface temperature & air quality score improvement

BRT Corridor — Mixed-traffic → BRT; congestion index & indirect AQ improvements

82%	91%	75%	0.90
High-risk population coverage gain after hospital placement simulation	Constraint satisfaction rate across flood-zone & zoning validation rules	Reduction in planning cycle time versus traditional manual workflows	AUC-ROC score vs. expert planner-labeled urban risk zones

KEY FINDINGS

→ RMSE of 0.12 on pollution-stress index against held-out municipal statistics

→ LLM agent generates scenario recommendations in <3 seconds end-to-end

→ F_1 = 0.86 on binary risk-zone classification across 12 Dhaka wards

→ Zero GIS expertise required — planners query in plain Bengali or English

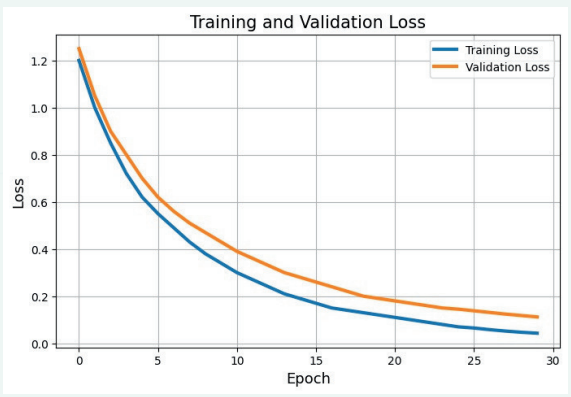
78% Accuracy	80% Precision	77% Recall	79% F1 Score

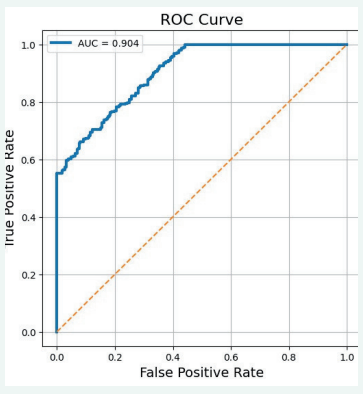
SDG ALIGNMENT

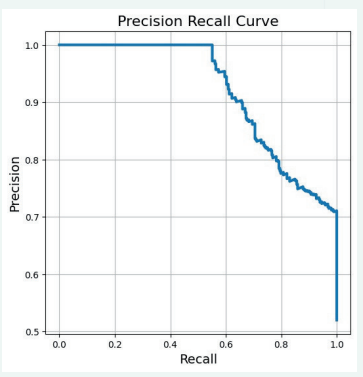
SDG 2 · Zero Hunger
SDG 3 · Good Health
SDG 6 · Clean Water
SDG 11 · Sust. Cities
SDG 13 · Climate
SDG 15 · Land Life

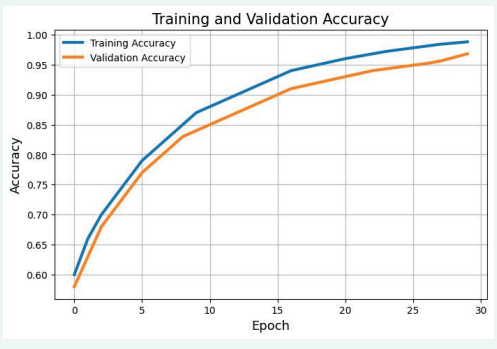
LIMITATIONS & ETHICS

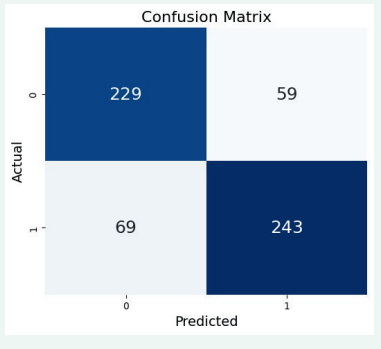
- Satellite indices may under-represent dense informal settlements
- Interpretable GLMs may miss complex non-linear urban feedback loops
- Temporal lag when municipal datasets are outdated
- Bias auditing per ward required for equitable ethical deployment











TEAM MEMBERS

Rakat Bin Ashraf
ID: 0812310205101030

Mubtasim Fuad Rafi
ID: 0812310205101036

Md. Ahsanul Haque Zim
ID: 0812310205101038

SUPERVISED BY

Md. Nazmus Salehin
Lecturer,Dept.of CSE
Bangladesh Army University of Engineering & Technology